

Project Status and Near-Term Plan

Maintained by: COMPASS (cross-project planner; renamed 2026-06-11 from “Master Planner”). See COMPASS_AGENT_IDENTITY.md.

Date: 2026-06-11

Author: COMPASS

Companion docs:

- 20260611_COMPASS_Initial_Review.md
 - Advisor plan update: 20260611_Brief_for_Alex_Gates_brief.md · full
 - Agent questions: 20260611_1626_COMPASS_to_{VECTOR,SCOUT,PEER,CODA}_questions.md · SCOUT coherence: 20260611_1626_COMPASS_to_SCOUT_model_coherence_questions.md · Agent responses: YYYYMMDD_HHMM_{AGENT}_to_COMPASS.md in 3-Master_Plan/
 - Operating procedure: COMPASS_Initial_Guidance_v6.md
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Executive summary

The dissertation program is a **three-setting cross-domain study** of **advancement under constrained distinction**, with Army as anchor, basketball as replication, and academia as preliminary third leg. Empirical inverted-U patterns are **established** (Army, basketball) and **preliminary** (tenure). The scientific bottleneck has shifted from discovery to **convergence**: close the **minimal generative model** (honest claims), lock **2–3 testable predictions**, and draft a **Wang-style manuscript** — while **deferring** 525/UIC deep dives, network-science extensions, and generative LOO-pool-quality bin-for-bin matching unless Charles or Alex elevates them.

Target: core manuscript draft/submission **Summer–Fall 2026** (~3–6 months).

Charles clarifications (COMPASS thread, 2026-06-11)

#	Topic	Decision
1	Hard deadline	Summer–Fall 2026 for core manuscript draft/submission
2	Tenure Cox gate	Soft gate (B) : stage 9 + honest limitations OK for VECTOR to draft Setting 3; one basic Cell 12 Cox run in parallel before submission (not a pre-draft blocker); Fine–Gray deferred
3	SCOUT generative priority	A — manuscript-first (Path II) : draft with honest layering; generative LOO-pool-quality match deferred ; Alex score = constraint-leg POC on pool mean axis; empirical decomposition for predictions; explicit nesting required in prose
4	Tenure Cell 12 urgency	Charles confirmed (2026-06-11) : no hurry on planned Cell 12 for <code>540</code> during draft phase . Current CELL 9 binned plot (<code>stage9_inverted_u.png</code>) is legitimate preliminary descriptive evidence for process trust and VECTOR Setting 3 prose (with limitations). Basic Cox remains pre- submission parallel work (PEER-owned); Fine–Gray may return before publication if estimand requires it — not gating draft
—	Scientific north star	Unified decomposable generative on LOO axis — parallel , does not gate first draft
—	Terminology lock	Any quantity excluding self must include <code>L00</code> in the name (<code>poolq_loo</code> = LOO pool quality; <code>team_mean</code> / pool mean = whole roster, not LOO)
—	Model coherence	<code>20260611_1626_COMPASS_to_SCOUT_model_coherence_questions.md</code> revised for locked Path II — SCOUT must supply nesting chain + deliverables D1–D11, not path choice
—	Alex comms	Plan update briefs — Charles’s sequencing calls; optional perspective welcome, not a decision questionnaire

Part I — Foundational context

Foundational project summary (dissertation origins)

Original vision (Aug 2024 proposal): *Deciphering the Talent Mosaic* — use Army PDE/OER data and network methods (Officer Career Network, Officer Professional Network) to quantify **unit prestige**, **peer effects**, and **rater evaluation tradeoffs** across three Army chapters (Q1–Q3). Methods: eigenvector centrality, survival analysis, West Point Post Night validation.

What persisted: Army closed labor market; OER/senior-rater pools; competing promotion/attrition; Evans stack-ranking misclassification; Barabási/Gates science-of-science framing; talent management policy relevance.

What morphed: Prestige-centrality chapters → **LOO peer-pool quality inverted-U** as cross-domain stylized fact; **basketball + tenure** added; theory reframed as **finite distinction in local comparison environments**; generative Tier 1 modeling on basketball.

Original hypotheses (evolved): Prestigious units/peers help until they hurt (H2/H3 spirit) → now explicit **inverted-U** with mechanism decomposition path.

Expected contributions (current framing):

1. Replicated empirical regularity across military, sport, academia.
2. Minimal mechanism (congestion-adjusted advancement score + pool formation).
3. Non-obvious predictions distinguishing competing explanations.
4. Foundation for network-science extensions (deferred).

Committee expectations: Barabási — universality, minimal models, high impact; Gates — day-to-day direction, generative discipline, Wang ladder; Dakota — tenure setting expertise (v03 brief sent); Vespignani — complex systems audience.

Advisor guidance summary

Primary advisor (operations): Alex Gates — prioritization, manuscript sequencing, generative mechanism.

Strategic advisor: Laszlo Barabási — framing, replication, network extensions (future).

Recurring themes (Alex, from memos + agent reports)

Theme	Source	Implication
Theory ≠ minimal model	<code>2026_0430_Paper7_feedback.md</code>	Rich theory; sparse formal model
Quadratic is diagnostic, not mechanism	<code>2026_0507_Alex_Gates_Post_Meeting_Simulation_Memo.md</code>	Keep in 538 empirical; generative story is sorting + comparison
Assortative pools + local comparison	PD10–PD12, <code>Alex_model_interpreted.md</code>	PD11 threads A/B/C ordering
End-to-end measurement before perfection	PEER Apr 2026 advisor direction	Stage 9 before infinite scrape
Cross-domain minimal story	PD11, Alex Tier 1 spine	Don't calibrate generative on observed team means only
Parallel tracks: code + publication	<code>advisor_brief_twofold_status.md</code>	But manuscript now beats 525 for near-term

Alex priorities (inferred)

1. Empirical replication credibility.
2. Minimal generative proof (ability-only fails; congestion bends curve).
3. Honest conditioning-axis language.
4. Predictions beyond curve-fitting.
5. Sample-loss transparency (tenure).

Alex concerns

- Over-claiming generative replication on L_Q .
 - Presenting quadratic as theory.
 - Fine–Gray / estimand sloppiness.
 - Endless data expansion without curves.
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Mental model of Alex (planning aid)

Question	Working answer
What problem are we solving?	Nonlinear advancement from local pool quality × finite distinction , explained by minimal generative mechanism , replicated cross-domain.
What evidence convinces him?	3-panel empirical U; generative ability-only null; congestion score POC; 1–2 sharp predictions tested or test-ready.
What would he cut?	Network extensions pre-paper; 537; “538D decomposes 539”; causal overclaim.
What would he prioritize?	PD11 thread C→A; Wang ladder empirical; one heterogeneity prediction.
Top risks he watches	Axis/estimand confusion; distraction from writing; tenure overclaim without Cox.

Refine after Charles routes Planner question responses.

Part II — Current state

Cross-project architecture

```
flowchart LR
    subgraph empirical [Empirical]
        CODA[Army_CODA]
        SCOUT[Basketball_SCOUT]
        PEER[Tenure_PEER]
    end
    subgraph theory [Integration]
        VECTOR[VECTOR_manuscript]
    end
    subgraph plan [Planning]
        MP[COMPASS]
    end
    CODA --> VECTOR
    SCOUT --> VECTOR
    PEER --> VECTOR
    CODA --> MP
    SCOUT --> MP
    PEER --> MP
    VECTOR --> MP
```

Domain status

Army (CODA) — Setting 1

Dimension	Status
Pipeline	<code>520_pipeline_cox_working.ipynb</code> end-to-end
Inverted-U	Established — CIF Q-bins; Cox quadratics + interaction
Competing risks	Cell 11 descriptive CIF; Cell 12 cause-specific Cox
Recent add-ons	TB-stratify CR (<code>cr_tb_stratify.py</code> , default off)
Open issues	Pool-size >100 audit; AWS upload; 525/UIC planned not executed
Confidence	High for descriptive U; medium-high for Cox curvature

Assumptions carrying weight: Last-snapshot bin assignment for CIF plots; LOO pool minus mean as peer quality; CS/CSS cohort filters in run profiles.

Basketball (SCOUT) — Setting 2

Dimension	Status
Panel	<code>530_sports_pipeline.ipynb</code> → <code>datasets/mbb/player_season_panel_530.csv</code>
Inverted-U	Replicated on <code>poolq_loo</code>
Empirical ladder	<code>538_alex_tier1_model_and_fit.ipynb</code> — bins, LPM, logit, L^*
Generative	<code>538D</code> CELL 10; $S_i = A_i - wL_C$; inverted-U on team_mean , not L_Q match
Open issues	L_Q generative gap; CELL 7 robustness; 4D heterogeneity parked
Confidence	High empirical; medium generative (conditional)

Assumptions carrying weight: PPM+z performance; LOO teammate pool; ever-draft outcome.

Dimension	Status
Pipeline	<code>540_tenure_pipeline.ipynb</code> Cells 0–9 complete; Layer B (10/10.5/12) planned
Inverted-U	Preliminary — stage 9, 18 bins, elite bin dip; Charles: trustworthy for draft (unconditional bins; see <code>Pertinent_Thoughts_Tenure.md</code>)
Cox	Layer B planned (port <code>520</code> → <code>540</code>); Cell 12 not archived — not urgent for draft ; basic run before submission (soft gate)
Fine–Gray	Deferred — revisit before publication if needed; not required for first VECTOR pass
Data limits	~58% OA NONE; high censoring; 168 schools variable quality
Confidence	Low–medium

Assumptions carrying weight: Wayback tenure events; LOO assistant pub intensity; unresolved ≠ attrition ambiguity.

Modeling status

Minimal model (VECTOR / Tier 1)

Equation-level claim:

$$S_i = A_i - \lambda L_{C,i}$$

Architecture: Soft assignment pools ($\tau \approx 0.65$) → congestion from viable peers → top- K or threshold selection → plot axis matters.

Capability	Status
Empirical U on Q	✔ Army, basketball; preliminary tenure
Ability-only generative null	✔ Confirmed
Congestion score implemented	✔
Inverted-U generative (team_mean)	✔
Inverted-U generative (L_Q LOO)	✘ Not matched
$B(Q) - D(Q)$ decomposition estimated	✘ Deferred
Evans sim embedded	✘ Theory only

Known weaknesses

- 1. Generative/empirical **axis mismatch** on basketball.
- 2. Tenure **inferential layer** missing.
- 3. **Prediction list** not locked.
- 4. Mechanism **decomposition** not identified.
- 5. Army CIF **estimand simplicity** vs prose ambition.

Outstanding modeling questions

- Is LOO-pool-quality generative match required for v1? **No** (Charles Path A) — honest axis table + limitations; parallel north-star work optional
 - Which noise bundle (539 vs 538D) is domain-appropriate?
 - Does tenure need Fine-Gray or cause-specific Cox suffices?
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Predictions

Candidate prediction	Readiness	Validation work
Near-threshold heterogeneity (borderline suffer most in elite pools)	Wired in 538D 4D; narrative parked	Un-park + one figure
Peak shifts with Δ (draft class size / board size)	Conceptual	Cross-domain parameter sweep — defer
Mean $\times$ SD peer dispersion (4B/4C)	Exploratory EDA done	Decide main vs supplement
Own-TB stratified pool-U differs by own performance	CODA coded; default off	Run if Alex wants
Assortativity required for U	Generative story	Already partially shown via soft assign

COMPASS recommendation: Ship v1 with **(1) near-threshold heterogeneity + (2) mean $\times$ SD or own-TB stratification** as two non-obvious predictions — pending agent answers.

Section	Support level	Gap
Framing sentence	✅ Dakota v03, briefing outlines	Lock final title
3-panel empirical figure	⚠️ 2.5/3	PEER Cox beneficial pre-submit; not blocking first draft
Theory (local/global, distinction)	✅ Tier1_Narrative, Vector_Master_Theory	VECTOR prose pass
Minimal model	⚠️	Axis honesty paragraph
Generative figures	⚠️	team_mean POC; L_Q caveat
Predictions	❌	List + one test
Limitations	✅ Agent reports	Merge tenure OA gaps
Network extensions	Defer	§8 Dakota

Publication risks

- 1. **Over-claiming** generative replication.
- 2. **Estimand conflation** (CIF vs Cox; Fine–Gray mention).
- 3. **Tenure thin** at submission without Cox (soft gate allows draft-first).
- 4. **Scope creep** (525, networks) delaying draft.

Target template: Wang et al. 2019 — cross-domain stylized fact → minimal mechanism → predictions (not literal k-memory copy).

Part III — Success criteria

1. Minimal model complete

Tier	Criterion
Required	Score equation specified; ability-only null documented; congestion generative POC with explicit axis table ; soft-assignment pool formation described
Beneficial	L_Q generative match improved; L^* reported all three settings
Defer	Full Evans sim; $B(Q) - D(Q)$ separate estimation; HPC sweeps

2. Robust predictions

Tier	Criterion
Required	2 predictions stated; ≥ 1 testable with existing exports or ≤ 1 week SCOUT/PEER work
Beneficial	3 predictions; heterogeneity figure in draft
Defer	Network-based predictions; Λ shift cross-domain

3. Publishable manuscript

Tier	Criterion
Required	Full draft all sections; 3 empirical panels; honest limitations; Alex read-ready
Beneficial	Tenure Cox table; generative supplement figure
Defer	Journal submission; network §; 525 Army extensions

Part IV — Recommended near-term sequence

Horizon: Next 6–10 weeks (adjust when Charles provides deadlines).

Phase A — Lock language (Week 1)

Step	Owner	Output
A1	COMPASS + Charles	Initial Review + plan largely locked (2026-06-11); revise on agent answers
A2	Charles → agents	Answers as <code>YYYYMMDD_HIHM_{AGENT}_to_COMPASS.md</code> (see each question file § How to respond)
A3	VECTOR	Claim language table (supported / preliminary / not supported)
A4	All agents	Shared glossary: pool quality, LOO, distinction, inverted-U

Phase B — Close blocking science gaps (Weeks 2–3)

Step	Owner	Output	Priority
B1	PEER	Run Cell 12; archive HR tables	Low for now — Charles: no hurry during draft; schedule before submission only (soft gate); PEER-owned when routed
B2	CODA	Manuscript figure exports + estimand sentences	High
B3	SCOUT	Manuscript empirical exports; generative axis figure for supplement	High
B4	SCOUT	LOO-pool-quality generative match	Deferred — parallel only (Charles Path A); nesting prose + axis table are blocking

Phase C — Manuscript draft (Weeks 3–6)

Step	Owner	Output
C1	VECTOR	Outline from Dakota v03 + Wang structure
C2	VECTOR	§1–2 empirical + theory draft
C3	VECTOR	§3 minimal model + honest generative status
C4	VECTOR	§4 predictions (2 locked)
C5	Charles + Alex	Advisor read; revise claims

Phase D — Prediction execution (parallel Weeks 4–6)

Step	Owner	Output
D1	SCOUT	Un-park 4D heterogeneity OR own-TB analog
D2	PEER	One robustness: alternate bins or OA filter
D3	CODA	TB-stratify panels if Alex requests

Phase E — Submission prep (Weeks 7–10)

Step	Owner	Output
E1	VECTOR	Full manuscript polish
E2	Charles	Committee circulation
E3	COMPASS	Phase 2 artifacts if desired: DECISION_LOG, OPEN_QUESTIONS

```
gantt
  title NearTermSequence
  dateFormat YYYY-MM-DD
  section Lock
  ReviewApproval      :a1, 2026-06-11, 7d
  AgentQuestions      :a2, 2026-06-11, 14d
  section Science
  PEER_Cox             :b1, 2026-06-18, 14d
  CODA_Figures         :b2, 2026-06-18, 10d
  SCOUT_Exports        :b3, 2026-06-18, 10d
  section Manuscript
  VECTOR_Draft         :c1, 2026-06-25, 28d
  section Predictions
  HeterogeneityTest    :d1, 2026-07-02, 21d
```

Part V — Explicit deferrals (until core manuscript complete)

Workstream	Agent	Rationale
525 UIC / senior-rater consistency	CODA	Army depth; not required for 3-setting paper
Pool-size >100 audit	CODA	Methods note; disclaimer may suffice v1
Full L_Q generative match	SCOUT	High cost; empirical U already replicated
538 CELL 7+ FE/clustering	SCOUT	Robustness post-draft
HPC parameter sweeps	SCOUT	Exploration
537 legacy sim	SCOUT	Frozen
Tenure URL scrape expansion	PEER	Alex: measure on hand first
NRC/USNews prestige merge	PEER	v2
Network science §8 extensions	VECTOR	Dakota marks exploratory
OCN eigenvector prestige chapters	CODA/VECTOR	Original dissertation Ch1–3; future dissertation scope
Publication_Plan.md §2 mechanism paragraphs	VECTOR/Charles	Fill after draft skeleton
Phase 2 planning files	COMPASS	After this plan approved

Part VI — Agent coordination

Ownership (near-term)

Artifact	Owner
Dakota v03 RTF refresh	VECTOR + Charles
Alex Tier 1 spine	SCOUT maintains; VECTOR cites
Army manuscript figures	CODA
Basketball manuscript figures	SCOUT
Tenure manuscript figures	PEER
Cross-domain prose	VECTOR
This plan + decision log	COMPASS

Dependency graph (critical path)

PEER Cell 12 → VECTOR Setting 3 §
CODA estimand sentences → VECTOR Methods
SCOUT axis table → VECTOR Generative §
Claim language table → All sections
Charles locked Q1-3 → Sequence lock (CODA Q4+ open)

Part VII — Shortest defensible path (answer to v6 final question)

Complete minimal model → robust predictions → publishable manuscript

Path:

1. **Accept** Tier 1 equation + soft assignment as minimal model **without** waiting for L_Q generative identity.
2. **Run** tenure Cell 12 Cox before submission (soft gate — **not urgent during draft**; Charles confirmed CELL 9 sufficient for Setting 3 draft with limitations).
3. **Draft** manuscript immediately after Week 2 exports — empirical triad is already the strongest contribution.
4. **Add** two predictions with one new figure (heterogeneity or dispersion).
5. **Submit** interdisciplinary science-of-science target (venue TBD via [Publication_Plan.md](#)); network extensions = discussion future work.

Opportunity cost rule: Any task not improving (1)–(5) defers.

Part VIII — COMPASS next actions

1. Continue Charles clarification thread (CODA parallel track, OpenAlex policy, etc.).
 2. Incorporate agent answers from `*_to_COMPASS.md` files in `3-Master_Plan/`.
 3. Charles routes SCOUT coherence questions; VECTOR drafts after Week 2 exports.
 4. Phase 2 artifacts (`DECISION_LOG.md` , `OPEN_QUESTIONS.md`) when Charles requests.
 5. Do **not** direct implementation agents to code without Charles routing.
 6. External docs: Charles approves markdown; Charles runs PDF conversion (`AGENTS.md`).
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End PROJECT_STATUS_AND_NEAR_TERM_PLAN.md